

Project Design Phase-II

Data Flow Diagram & User Stories

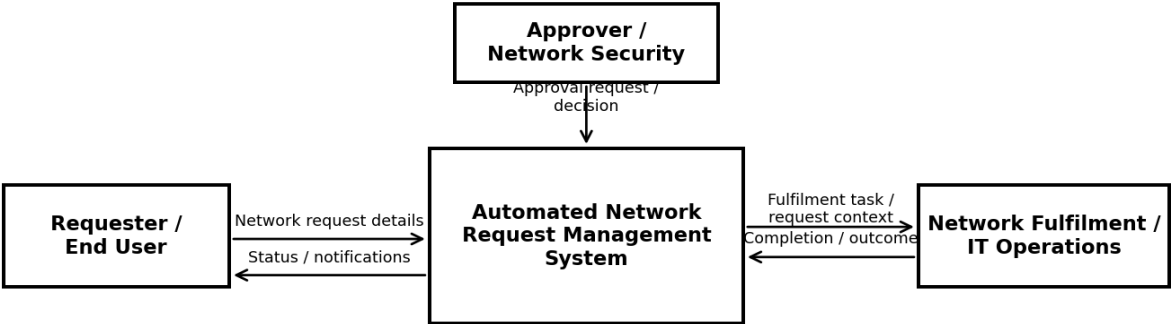
Date	02 October 2026
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Data Flow Diagrams:

The DFD represents how network request information moves through the ServiceNow solution: requester inputs enter through the Service Catalog, catalog variables are retrieved and mapped, Flow Designer creates the operational record, approval is requested and evaluated, the record is updated, notifications are sent, and the final request outcome is made available for fulfilment and tracking.

DFD Level 0 - Context Diagram

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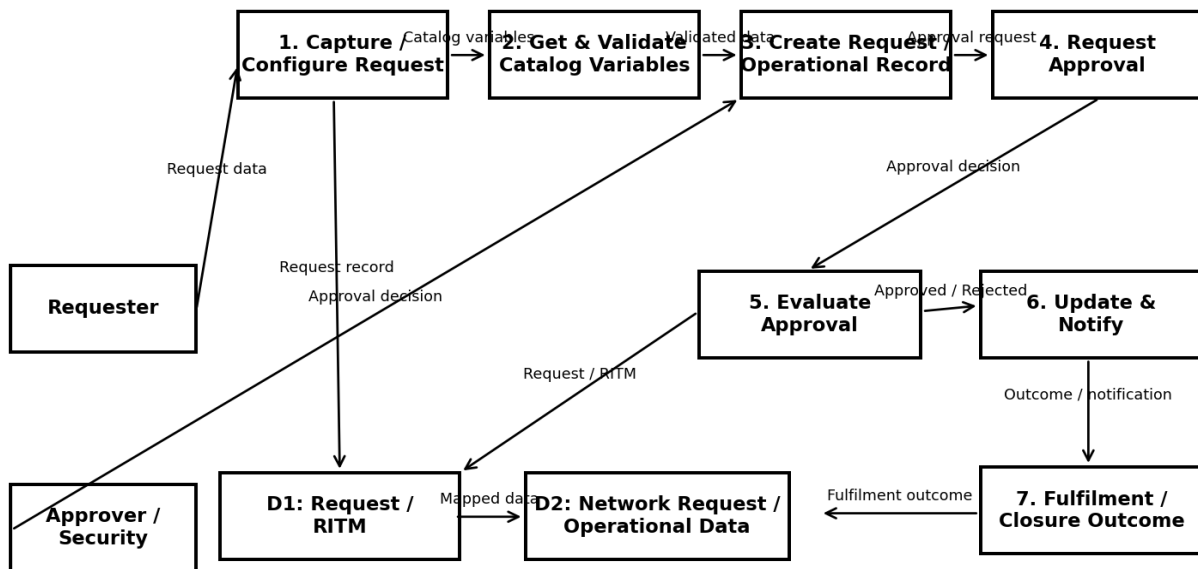


Centralized request, approval, fulfilment, notification and audit flow

Level 0 treats the Automated Network Request Management System as one process. The requester supplies network-request details and receives status/notification outputs. Approvers or network-security personnel provide approval decisions, while the fulfilment/IT operations side receives the approved request context and returns the operational outcome.

DFD Level 1 - Detailed Data Flow

DFD Level 1 - Detailed Network Request Data Flow



ServiceNow backbone: Service Catalog → Flow Designer → Record → Approval → Condition → Update → Notification

Level 1 Process Description

No.	Process	Data Flow / Function
1	Capture / Configure Request	Collect network-request information through the Service Catalog and provide the submitted Requested Item Record to the flow.
2	Get & Validate Catalog Variables	Retrieve the selected variables from the Network Request catalog template and make them available to downstream actions.
3	Create Request / Operational Record	Create the configured operational record and map catalog values into its fields; initialize Approval Status as Requested.
4	Request Approval	Send the created record into the configured approval action and obtain an approval decision.
5	Evaluate Approval	Check the approval output and branch the flow according to whether the request is Approved or follows the Else path.
6	Update & Notify	Update the operational record with the decision and send the corresponding lifecycle notification.
7	Fulfilment / Closure Outcome	Pass the approved/rejected outcome into the operational lifecycle for fulfilment, tracking and closure.

Data Stores

- **D1 - Request / RITM:** Maintains the ServiceNow request and request-item context created when the requester submits the catalog item.
- **D2 - Network Request / Operational Data:** Represents the structured record populated by the flow. The supplied implementation screenshots currently show the visible table label 'Database Tables'; this terminology is retained as implementation evidence.

User Stories

The user stories below translate the implemented and planned functional requirements into user-centered tasks. Priority and release values are organized around the project implementation roadmap and distinguish current workflow capabilities from future enhancements.

User Type	Functional Requirement (Epic)	User Story No.	User Story / Task	Acceptance Criteria	Priority	Release
Requester	Network Request Submission	USN-1	As a requester, I can open the Network Request catalog item and enter my required request information.	The catalog item opens and the required request variables can be entered successfully.	High	Phase 1
Requester	Catalog Variable Capture	USN-2	As a requester, I can submit structured network-request data through the ServiceNow portal.	The Requested Item Record is created and the configured flow receives the submission context.	High	Phase 1
System	Variable Retrieval	USN-3	As the system, I can retrieve the selected catalog variables from the Network Request item.	The configured variables are available as outputs for downstream flow actions.	High	Phase 2
System	Operational Record Creation	USN-4	As the system, I can create an operational record using the submitted request values.	Mapped fields are populated correctly and Approval Status starts as Requested.	High	Phase 2
Approver	Request Approval	USN-5	As an approver, I can review the network request and provide an approval decision.	The configured approver receives the request and an approval outcome is recorded.	High	Phase 2
System	Approval Routing	USN-6	As the system, I can evaluate the approval result and follow the correct workflow branch.	Approved requests follow the Approved branch; other outcomes follow the Else path.	High	Phase 2
Requester	Request Notification	USN-7	As a requester, I receive a notification when my request has been created.	A request-created email is generated after the operational record is created.	High	Phase 2

User Type	Functional Requirement (Epic)	User Story No.	User Story / Task	Acceptance Criteria	Priority	Release
Requester	Resolution Notification	USN-8	As a requester, I receive a notification when my approved request reaches the configured resolved stage.	The approved path updates the record and sends the configured resolved notification.	High	Phase 3
Network / IT Operations	Operational Record	USN-9	As an operations user, I can access structured network-request records for fulfilment and tracking.	Mapped request information is stored in the configured operational record.	High	Phase 3
IT Administrator	Workflow Monitoring	USN-10	As an administrator, I can inspect Flow Designer execution details when a request fails.	Execution details identify the failed action, condition path, input or output involved.	High	Phase 4
Administrator	Access Control	USN-11	As an administrator, I can configure roles and ACLs to control access to network-request records.	Unauthorized users cannot access protected operational data beyond their permitted role.	High	Phase 4
Administrator	Maintainability	USN-12	As an administrator, I can maintain the workflow primarily through Flow Designer and ServiceNow configuration.	Core workflow logic can be understood and modified without unnecessary custom server-side code.	Medium	Phase 4
Requester	Status Tracking	USN-13	As a requester, I can understand the current status of my submitted network request.	The request status is available through the configured ServiceNow request experience.	Medium	Future
Network / IT Operations	Fulfilment Handoff	USN-14	As a fulfilment user, I can receive the approved request context needed to perform the network operation.	Approved request information is available to the responsible fulfilment process or task.	Medium	Future

Acceptance Focus

The critical acceptance path is: requester enters valid network-request information → request is submitted through the Service Catalog → catalog variables are retrieved → the operational record is created → approval is requested → the approval result is evaluated → the record is updated → the appropriate notification is sent → the resulting request is available for fulfilment and tracking. Debugging of variable mappings, record creation, approval routing, condition logic and notification triggers is therefore directly linked to the high-priority user stories.